Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/08/2018
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Project Title:	Quantitative Studies of Cool Flame Transitions at Radiation/Stretch Extinction Using Counterflow Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-A: Use of the NASA Physical Sciences Informatics System – Appendix A
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reuter, Christopher M.S. (Princeton University)		
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	Cool flames have significant implications for the control of engine knocking, ground and space fire safety, and combustion emissions. The recent International Space Station (ISS) combustion science experiments on flame extinguishment discovered the existence of cool flames for both large single droplets and droplet arrays after radiative extinction. Numerical simulations of droplet combustion also provide an explanation for the transition of hot flame to cool flame after radiative extinction. However, it remains unclear at what conditions the hot flames transfer to cool flames and how the cool flame flammable region is related to hot flame flammability limit and extinction limit. It is also not clear how oxygen concentration, dilution, pressure, diffusion, and convection transport processes affect the cool flame transition. More importantly, due to limits of space and payload, the ISS experiments did not provide detailed speciation and temperature distributions of cool flame structures for quantitative validation of kinetics and radiation		

Task Description:	model. By using the Flame Extinguishment (FLEX) experimental data from the Physical Sciences Informatics (PSI) system of combustion science, we propose to conduct new simulations and ground-based experiments using ozone-sensitized counterflow flames to * develop a new diagram of the flammability limits of diffusional cool flames affected by radiation, diffusion, and convection ; * bridge the knowledge gap between the flammability limits and extinction limits of hot flames and cool flames ; * predict and answer the question at what conditions the transition from hot flame extinction to cool flames is possible and verify the phenomena observed on the ISS flame extinguishment experiments ; * obtain new, quantitative data of extinction limit, flame temperature, and species distribution for the validation of kinetic and radiation models involving cool flames ; * extend the cool flame experiments from the original ISS droplet experiments to a more general ground-based combustion system involving radiation, diffusion, convection, and low-temperature chemistry while maintaining a one-dimensional geometry. The proposed new ground-based experiments and data analysis are complimentary to and not directly linked with the science objectives from the original ISS combustion science investigation, and will significantly advance the understanding of near-limit cool flame dynamics and contribute to the future control of engine knocking, ground and space fire safety, and combustion emissions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Numerical calculations of dimethyl ether (DME) radiative flames in the counterflow geometry found three distinct flame regimes at elevated pressures—hot flames, warm flames, and cool flames. The results showed that the warm flame has a double reaction zone structure and plays a critical role in the transition between cool flames and hot flames. At elevated pressures, it was found that the flammable region of warm flames and cool flames expands due to the increase in low-temperature and intermediate-temperature reactivity with pressure. Since many real engines operate in the 20-40 atm pressure range, therefore, cool flames and warm flames are relevant and important flame regimes for high-pressure near-limit combustion. These calculations served to advance the understanding of near-limit flame dynamics and provide guidance for experimental observation of different flame regimes. The warm diffusion flame was then revealed to occur in counterflow burner experiments using dibutyl ether (DBE). The two-stage flame structure was confirmed. Therefore, warm flames serve as a bridge between cool flames and hot flames. The warm diffusion flame is particularly relevant to combustion in highly turbulent flow fields and in microgravity environments, owing to the possibility of long residence times.
Task Progress:	We made two main progresses in Year 2 of our research: (1) we performed numerical calculations of dimethyl ether (DME) radiative flames in the counterflow geometry, discovering the “warm flame” in the process, and (2) we confirmed these warm flames experimentally using dibutyl ether (DBE). While recent calculations have shown that cool premixed flames can extend the conventional lean and rich flammability limits of the hot flame, the flame regimes and burning limits of diffusion flames have not been as thoroughly investigated, especially at increased pressures. Therefore, the objective of the first study was to investigate the extinction and ignition transitions between different diffusion flame regimes at elevated pressures and to obtain a general diffusion flame flammability diagram which includes both hot and cool flames. Specifically, the impacts of radiation, pressure, and reactant concentration on the extinction limits of different DME flame regimes were examined computationally with and without thermal radiation. A general Є-shaped flammability limit diagram for diffusion flames was then obtained. In addition, the existence of a DME warm flame regime with a double reaction zone structure was demonstrated, and its flame structure and chemistry were compared with that of cool and hot flames. It was shown that there are three possible cool flame transitions: (1) radiative extinction of hot flame to cool flame, (2) stretch extinction of hot flame to cool flame, and (3) stretch extinction of warm flame to cool flame. Moreover, there are also two different ways to form a warm flame by (1) radiative extinction of a hot flame and (2) ignition of a cool flame. The identification of these three flame regimes helps to understand near-limit flame transitions and to guide experimental observation of warm and cool flames. In the second study, we reported on experimental evidence of the existence of a new self-sustaining low-temperature multistage warm diffusion flame, existing between the cool flame and hot flame, at atmospheric pressure in the counterflow geometry. The structure of multistage DBE warm diffusion flames was examined by using thermometry, laser-induced fluorescence, and chemiluminescence measurements. It was confirmed that the warm diffusion flame has a two-staged double flame structure. A leading diffusion cool flame stage is present on the fuel side with a second intermediate stage on the oxidizer side. The second stage has a strong heat release that can be comparable to that of the first stage. Furthermore, the analysis revealed that with the increase of fuel concentration, a single-stage cool flame can ignite to a warm flame or a hot flame. Moreover, a warm flame can extinguish into a cool flame or ignite to a hot flame when the fuel concentration is substantially reduced or increased, respectively. Finally, under certain conditions, a hot flame can extinguish directly into either a warm flame or a cool flame. Hence, the results suggest that the multistage warm flame can act as a critical bridge between cool flames and hot flames. The following have been submitted to Proceedings of the Combustion Institute: Lin E, Reuter CB, Ju Y. "Dynamics and Burning Limits of Near-Limit Hot, Mild, and Cool Diffusion Flames of Dimethyl Ether at Elevated Pressures" Proceedings of the Combustion Institute 37 O. R. Yehia, C. B. Reuter, Y. Ju "On the Chemical Characteristics and Dynamics of n-Alkane Low-temperature Multistage Diffusion Flames" 37.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Yehia OR, Reuter CB, Ju Y. "Low-temperature multistage warm diffusion flames." Combustion and Flame. 2018 Sep. 195:63-74. https://doi.org/10.1016/j.combustflame.2018.03.013 , Sep-2018
Papers from Meeting Proceedings	Reuter CB, Lin E, Ju Y. "Pressure and Radiation Effects on the Dynamics of Hot and Cool Diffusion Flames." 26th International Colloquium on the Dynamics of Explosions and Reactive Systems, Boston, MA, July 30-August 4, 2017. ICDERS-1073, 26th International Colloquium on the Dynamics of Explosions and Reactive Systems, Boston, MA, July 30-August 4, 2017. , Aug-2017